

ASSIGNMENT CLA-01

Principles of Signal Processing (EEE310)

Signals and Systems – Analytical Problems

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TABLE OF CONTENT

S.NO.	CONTENT	PAGE NO.
1	Section A [A1 to A6]	2
2	Section B [B1 to B4]	2
3	Section C [C1 to C3]	2-3
4	Section D [D1 to D4]	4
5	Section E [E1 to E4]	4

Section-A

Question A1:

For

$$x(t) = \cos(12\pi t) + \sin(18\pi t), \quad y(t) = x(2t - 1),$$

the fundamental period of $y(t)$ is $\frac{1}{6}$ s. Establish the actual fundamental period from first principles; identifying a common period that is not the smallest earns no credit in self-evaluation.

Answer:

A1: $x(t) = \cos(12\pi t) + \sin(18\pi t), \quad y(t) = x(2t - 1)$

Sol: $y(t) = \cos[12\pi(2t - 1)] + \sin[18\pi(2t - 1)]$

$$= \cos[24\pi t - 12\pi] + \sin[36\pi t - 18\pi] \quad (1)$$

$$= \cos(24\pi t) + \sin(36\pi t) \quad (2)$$

A signal is periodic if there exists T (time period) > 0 i.e. $y(t + T) = y(t)$

Applying $y(t + T) = y(t)$ in the eq2 function given

$$\begin{aligned} y(t + T) &= \cos(24\pi(t + T)) + \sin(36\pi(t + T)) \\ &= \cos(24\pi t + 24\pi T) + \sin(36\pi t + 36\pi T) \end{aligned}$$

$\Rightarrow$ Cos part

$$24\pi(t + T) = 24\pi t + 2\pi$$

$$24\pi T = 2\pi$$

$$T = \frac{1}{12}$$

$\Rightarrow$ Sine part

Sine completes one full cycle θ by 2π

$$36\pi(t + T) = 36\pi t + 2\pi$$

$$36\pi T = 2\pi$$

$$T = \frac{1}{18}$$

$\Rightarrow$ Now for the total time period of $y(t)$ least common multiple of $\left[\frac{1}{12}, \frac{1}{18}\right]$

is $\frac{1}{6}$

$\therefore$ So the fundamental period of $y(t) = \frac{1}{6}$ s

Question A2:**Answer:**

$\Rightarrow x(t)$ is periodic with fundamental period T_0

$$x(t) = x(t + T_0) \quad \textcircled{1}$$

02 The energy of the $x(t)$

$$E_0 = \int_0^{T_0} |x(t)|^2 dt \quad E_0 > 0 \quad \textcircled{2}$$

03 Calculating total energy (for a continuous time)

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt \Rightarrow E = E_0 + E_0 + \dots = \infty$$

04 Since $E_0 > 0$

$$E = E_0 + E_0 + \dots > 0 \Rightarrow E > 0$$

$$\{E = \infty\} \quad \textcircled{3}$$

05 Average power of $x(t)$

$$P = \frac{1}{T_0} \int_0^{T_0} |x(t)|^2 dt \quad \textcircled{4}$$

$$\Rightarrow P = \frac{E_0}{T_0}$$

$$\Rightarrow P > 0 \text{ since } E_0 > 0 \text{ \& } T_0 > 0 \quad \textcircled{5}$$

Example: $x(t) = 2 \cos(4\pi t)$

$$T_0 = \frac{2\pi}{4\pi} = \frac{1}{2} s$$

$$\Rightarrow E_0 = \int_0^{1/2} |2 \cos(4\pi t)|^2 dt$$

$$= \int_0^{1/2} 4 \cos^2(4\pi t) dt$$

$$= 4 \left[\frac{t}{2} + \frac{\sin(8\pi t)}{8\pi} \right]_0^{1/2}$$

$$= 4 \left[\frac{1}{4} \right]$$

$$E_0 = 1$$

$$\Rightarrow E_{\text{total}} = 1 + 1 + \dots = \infty$$

$$\Rightarrow P = \frac{E_0}{T_0} = \frac{1}{1/2} = 2 > 0$$

Question A3:

Answer:

01. Average value is zero
given half wave symmetry

$$x\left(t + \frac{T_0}{2}\right) = -x(t)$$

$$\Rightarrow X = \frac{1}{T_0} \int_0^{T_0} x(t) dt$$

$$= \frac{1}{T_0} \left[\int_0^{T_0/2} x(t) dt + \int_{T_0/2}^{T_0} x(t) dt \right] \quad (1)$$

$\therefore$ Since it is a half wave symmetry

$$\int_0^{T_0/2} x(t) dt = - \int_{T_0/2}^{T_0} x(t) dt \quad (2)$$

$$\int_0^{T_0} x(t) dt = 0$$

$$\bar{x} = 0$$

Statement one is true

02. Average value of $-x(t)$ is zero.

Average the absolute value

$$X_a = \frac{1}{T_0} \int_0^{T_0} |x(t)| dt$$

$$\text{Here } |x(t)| = -x(t)$$

$$X_a = \frac{1}{T_0} \left[\int_0^{T_0/2} |x(t)| dt + \int_{T_0/2}^{T_0} |x(t)| dt \right]$$

$$\int_0^{T_0/2} x(t) dt \neq - \int_{T_0/2}^{T_0} x(t) dt \quad (3)$$

So $X_a \neq 0$

$X_a > 0$

Example:

$$x(t) = \begin{cases} 1 & 0 < t < 1 \\ -1 & 1 < t < 2 \end{cases}$$

$$X = \frac{1}{2} \left[\int_0^1 1 dt + \int_1^2 -1 dt \right]$$

$$= \frac{1}{2}(1 - 1) = 0$$

$$X_a = \frac{1}{2} \left[\int_0^1 |1| dt + \int_1^2 |-1| dt \right]$$

$$= \frac{1}{2}(1 + 1) = 1$$

Question A4:

Answer:

$\Rightarrow$ Given $x(t) = x_e(t) + x_o(t)$

We know

$$x_e(t) = \frac{x(t) + x(-t)}{2} \quad (1)$$

$$x_o(t) = \frac{x(t) - x(-t)}{2} \quad (2)$$

$$x_e(-t) = x_e(t) \quad (3)$$

$$x_o(-t) = -x_o(t) \quad (4)$$

$\Rightarrow$ Now

$$\langle x_e, x_o \rangle = \int_{-\infty}^{\infty} x_e(t) x_o(t) dt$$

$$\langle x_e, x_o \rangle = \text{even} \times \text{odd} = \text{odd}$$

$$\langle x_e, x_o \rangle = 0$$

Integral of an odd function over symmetric limits is 0

$\therefore$ Hence they are orthogonal.

$\Rightarrow$ Energy of $x(t)$

$$E_x = \int_{-\infty}^{\infty} |x_e + x_o|^2 dt$$

Question A5:

Answer:

Given $q(t) = r(t+1) - 2r(t) + r(t-1)$

$$r(t) = tu(t)$$

$$\Rightarrow q(t) =$$

$$\text{If } t < -1$$

$$r(t+1) = r(t) = r(t-1) = 0$$

$$q(t) = r(t+1) - 2r(t) + r(t-1)$$

$$= 0$$

$$2. \quad -1 < t < 0$$

$$r(t+1) = t+1$$

$$r(t) = 0$$

$$r(t - 1) = 0$$

$$q(t) = t + 1 + 2(0) + 0$$

$$= t + 1$$

$$3. \ 0 < t < 1$$

$$r(t + 1) = t + 1$$

$$r(t) = t$$

$$r(t - 1) = 0$$

$$q(t) = t + 1 - 2(t) + 0$$

$$= 1 - t$$

$$4. \ t > 1$$

$$r(t + 1) = t + 1$$

$$r(t) = t$$

$$r(t - 1) = t - 1$$

$$q(t) = t + 1 - 2(t) + (t - 1)$$

$$= 0$$

$$\Rightarrow q(t) = \begin{cases} 0 & t < -1 \\ t + 1 & -1 < t < 0 \\ 1 - t & 0 < t < 1 \\ 0 & t > 1 \end{cases}$$

first derivative

$$\frac{dq}{dt} = u(t+1) - 2u(t) + u(t-1)$$

Second derivative

$$\Rightarrow \frac{d^2q}{dt^2} = \delta(t+1) - 2\delta(t) + \delta(t-1)$$

Question A6:

Answer:

Let's take $k(t)$ fundamental period T_1

$J(t)$ fundamental period T_2

$$D(t) = k(t) + J(t)$$

$\Rightarrow D(t)$ is always periodic? $\Rightarrow$ False

The $D(t)$ guaranteed to be periodic if the periods of $k(t)$ & $J(t)$ is commensurate.

$$\frac{T_1}{T_2} \in \mathbb{Q}$$

$$\text{Q1: } T_1 = 2, \quad T_2 = 3 \Rightarrow \text{LCM}(2, 3) = 6 \quad \checkmark$$

$$\text{Q2: } T_1 = 1, \quad T_2 = \sqrt{2}$$

$\Rightarrow$ arbitrary real periods sum need not be periodic

$\Rightarrow$ Is the fundamental period always LCM? False

$\Rightarrow$ Even when T_1 and T_2 have common period the sum can have a smaller fundamental period than $\text{LCM}(T_1, T_2)$

Example:

$$k(t) = \sin(\pi t) + \sin(2\pi t)$$

$$\Rightarrow T_1 = 2, \quad T_2 = 1$$

$k(t)$ time period fundamental = 2

$$J(t) = -\sin(\pi t) + \sin(2\pi t)$$

$J(t)$ time period fundamental = 2

$$\Rightarrow \text{LCM}(T_k, T_J) = \text{LCM}(2, 2) = 2$$

Now add

$$\begin{aligned}
 k(t) + J(t) &= D(t) \\
 &= \sin \pi t + \sin 2\pi t \\
 &\quad + (-\sin \pi t) + \sin 2\pi t \\
 &= 2 \sin(2\pi t)
 \end{aligned}$$

Now

$$T_D = \frac{2\pi}{2\pi} = 1$$

$$\therefore \text{LCM} = 2 \quad \& \quad T_D = 1$$

Question B1:

Answer:

Section: B

$$x(t) = \begin{cases} -1, & -3 \leq t < -1 \\ 2t, & -1 \leq t < 1 \\ 3, & 1 \leq t < 2 \\ 0, & \text{otherwise} \end{cases}$$

$$y(t) = 1 - 2x(3 - 2t)$$

i) Sequence

$$x(3 - 2t) = x \left[-2 \left(t - \frac{3}{2} \right) \right]$$

ii) Time reversal: $x(t) \rightarrow x(-t)$

Time compression: $x(t) \rightarrow x(-2t)$

Right shift by $\frac{3}{2}$

$$x(t) \Rightarrow x(-2t) \Rightarrow x \left[-2 \left(t - \frac{3}{2} \right) \right]$$

and then

$$\text{multiply by } -2 \Rightarrow -2x \left[-2 \left(t - \frac{3}{2} \right) \right]$$

$$\text{shift amplitude by } 1 \Rightarrow 1 - 2x \left[-2 \left(t - \frac{3}{2} \right) \right]$$

$$\therefore y(t) = 1 - 2x(3 - 2t)$$

ii) Transformed break points

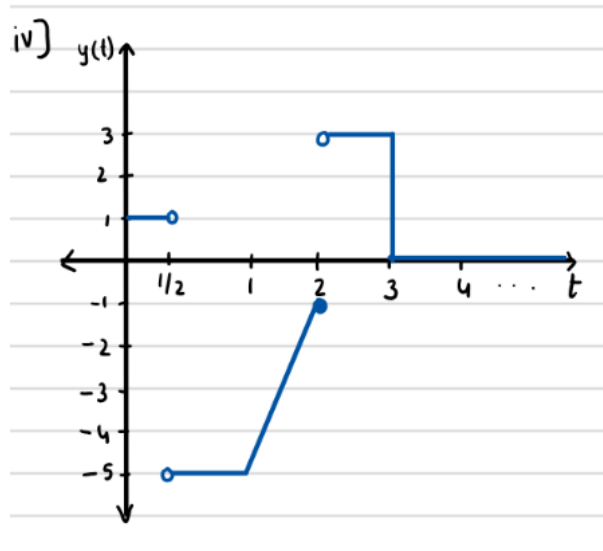


Figure 1: GraphB1

$x(t)$ original break points = $-3, -1, 2$

$\Rightarrow$ New break points

$$3 - 2t = -3 \Rightarrow t = 3$$

$$3 - 2t = -1 \Rightarrow t = 2$$

$$3 - 2t = 1 \Rightarrow t = 1$$

$$3 - 2t = 2 \Rightarrow t = \frac{1}{2}$$

New break points $0, \frac{1}{2}, 1, 2, 3$

iii)

$$y(t) = \begin{cases} -5, & \frac{1}{2} < t < 1 \\ 4t - 9, & 1 \leq t \leq 2 \\ 3, & 2 < t \leq 3 \\ 1, & t > 3 \end{cases}$$

Question B2:

For the same $x(t)$ in B1, derive $x_e(t)$ and $x_o(t)$ piecewise over the minimum set of intervals required. Sketch both and verify

$$x_e(-t) = x_e(t)$$

and

$$x_o(-t) = -x_o(t)$$

directly from your expressions.

Answer:

$$x(t) = \begin{cases} -1, & -3 \leq t < -1 \\ t + 2, & -1 \leq t < 1 \\ 3, & 1 \leq t < 2 \\ 0, & \text{otherwise} \end{cases}$$

For any signal,

$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

i) Find $x(-t)$

$$01) \quad x(t) = -1, \quad -3 \leq t < -1$$

$$\Rightarrow -3 \leq -t < -1 \Rightarrow 1 < t \leq 3$$

$$x(-t) = -1, \quad 1 < t \leq 3$$

$$02) \quad x(t) = t + 2, \quad -1 \leq t < 1$$

$$\Rightarrow -1 \leq -t < 1 \Rightarrow -1 < t \leq 1$$

$$x(-t) = 2 - t, \quad -1 < t \leq 1$$

$$03) \quad x(t) = 3, \quad 1 \leq t < 2$$

$$\Rightarrow 1 \leq -t < 2 \Rightarrow -2 < t \leq -1$$

$$x(-t) = 3, \quad -2 < t \leq -1$$

$$\therefore x(-t) = \begin{cases} 3, & -2 < t \leq -1 \\ 2 - t, & -1 < t \leq 1 \\ -1, & 1 < t \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

ii) Find $x_e(t)$

$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$\text{For } -3 \leq t \leq -2$$

$$x_e(t) = \frac{-1 + 0}{2} = -\frac{1}{2}$$

$$\text{For } -2 < t < -1$$

$$x_e(t) = \frac{-1 + 3}{2} = 1$$

$$\text{For } -1 \leq t \leq 1$$

$$x_e(t) = \frac{t + 2 + 2 - t}{2} = 2$$

$$\text{For } 1 < t < 2$$

$$x_e(t) = \frac{3 + (-1)}{2} = 1$$

$$\text{For } 2 \leq t \leq 3$$

$$x_e(t) = \frac{0 + (-1)}{2} = -\frac{1}{2}$$

$$\therefore x_e(t) = \begin{cases} -\frac{1}{2}, & -3 \leq t \leq -2 \\ 1, & -2 < t < -1 \\ 2, & -1 \leq t \leq 1 \\ 1, & 1 < t < 2 \\ -\frac{1}{2}, & 2 \leq t \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

iii) Find $x_o(t)$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

$$\text{For } -3 \leq t \leq -2$$

$$x_o(t) = \frac{-1 - 0}{2} = -\frac{1}{2}$$

$$\text{For } -2 < t < -1$$

$$x_o(t) = \frac{-1-3}{2} = -2$$

For $-1 < t < 1$

$$x_o(t) = \frac{t+2-(2-t)}{2} = t$$

For $1 < t < 2$

$$x_o(t) = \frac{3-(-1)}{2} = 2$$

For $2 \leq t \leq 3$

$$x_o(t) = \frac{0-(-1)}{2} = \frac{1}{2}$$

$$\therefore x_o(t) = \begin{cases} -\frac{1}{2}, & -3 \leq t \leq -2 \\ -2, & -2 < t < -1 \\ -1, & t = -1 \\ t, & -1 < t < 1 \\ 1, & t = 1 \\ 2, & 1 < t < 2 \\ \frac{1}{2}, & 2 \leq t \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

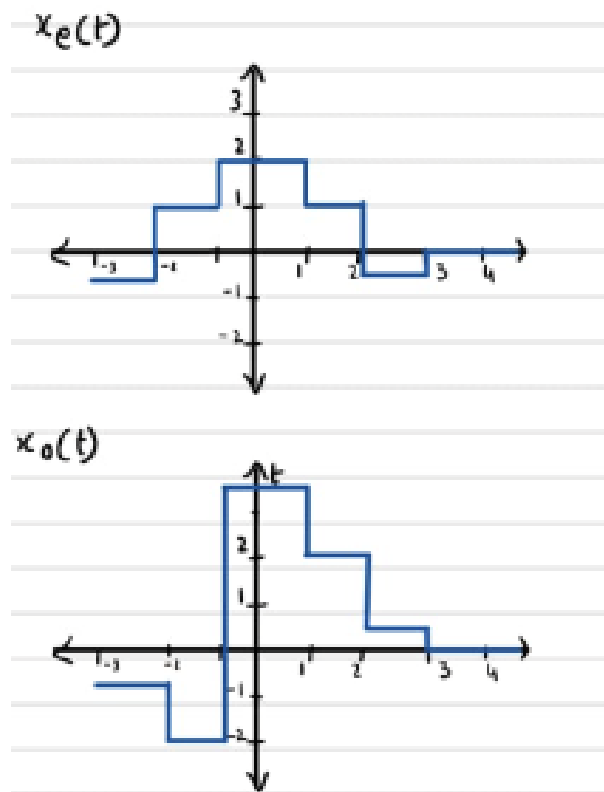


Figure 2: Graph B2

Question B3:

Answer:

$$x(t) = 2 \operatorname{rect}\left(\frac{t-1}{2}\right) - \operatorname{rect}\left(\frac{t+2}{4}\right)$$

$$y(t) = -3x\left(-\frac{t+1}{2}\right) + 2$$

Let us say,

$$u = -\frac{t+1}{2}$$

$$\Rightarrow x(t) = 2 \operatorname{rect}\left(\frac{t-1}{2}\right) - \operatorname{rect}\left(\frac{t+2}{4}\right)$$

$$\operatorname{rect}(t) = \begin{cases} 1, & -\frac{1}{2} < t < \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$$

$$-\frac{1}{2} < \frac{t+2}{4} < \frac{1}{2}$$

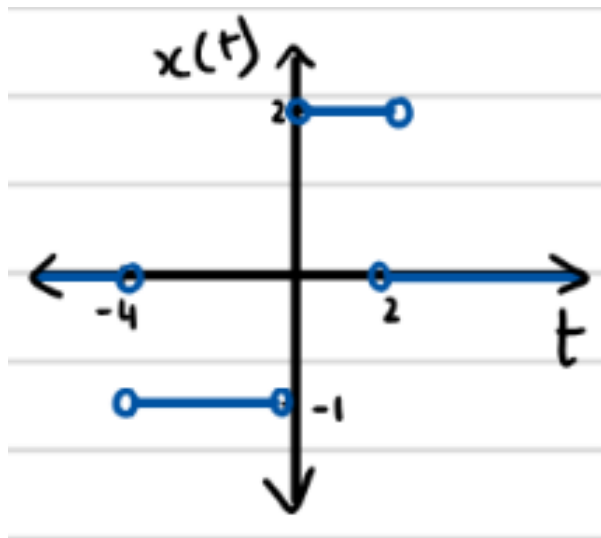


Figure 3: GraphB3A

$$\Rightarrow -2 < t + 2 < 2$$

$$\Rightarrow -4 < t < 0$$

So height = 2, $0 < t < 3$

$$2 \text{rect}\left(\frac{t-1}{2}\right)$$

$$\Rightarrow -1 < \frac{t-1}{2} < 1$$

$$\Rightarrow -2 < t - 1 < 2$$

$$\Rightarrow -1 < t < 3$$

Height = -1, $-4 < t < 0$

Elsewhere, 0

$$\Rightarrow \text{Now } y(t) = -3x\left(-\frac{t+1}{2}\right) + 2$$

Original $x(t)$

$$x(t) = \begin{cases} 2, & 0 < t < 3 \\ -1, & -4 < t < 0 \\ 0, & \text{elsewhere} \end{cases}$$

$$\text{Now } x\left(-\frac{t+1}{2}\right)$$

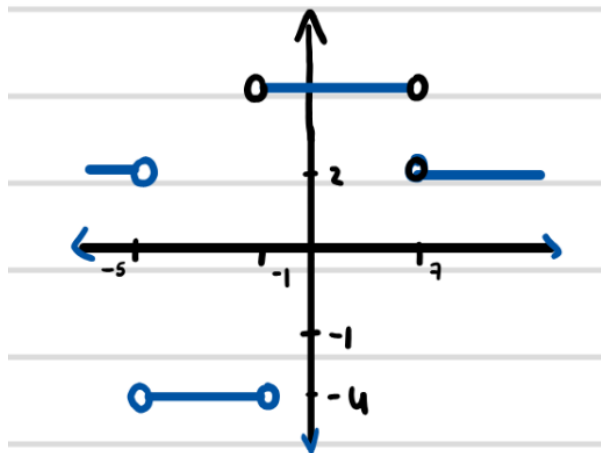


Figure 4: GraphB3B

$$-\frac{t+1}{2} = u$$

$$\Rightarrow x(u)$$

Range	Height
$0 < u < 3$	2
$-4 < u < 0$	-1
Elsewhere	0

Heights

$$-3x(u)$$

$$2(-3) = -6$$

$$-1(-3) = +3$$

$$+2 \rightarrow -6 + 2 = -4$$

$$-1 \rightarrow +3 + 2 = 5$$

$\therefore$ The u changes the width/location

while -3 and $+2$ change heights.

Question B4:

Answer:

$$x(-t) = -x(t)$$

$$x(t+4) = x(t)$$

$$x(t) = 0 \quad \text{for } |t| < 2$$

$\Rightarrow$ Odd signal from $x(-t) = -x(t)$

$$T = 4 \quad \text{from } x(t+4) = x(t)$$

$\Rightarrow$ Let us take one sec time limits $-2 < t < 2$

$$\text{If } x(t) = 0 \quad |t| < 2$$

$$\text{So } x(t) = 0 \quad \text{for } (-2, 2)$$

$$\Rightarrow x(t) = \begin{cases} -1, & -1 < t < 0 \\ 0, & t = 0 \\ 1, & 0 < t < 1 \\ 0, & 1 \leq t < 2 \end{cases}$$

$$\Rightarrow \int_{-2}^2 x(t) dt = 0$$

Since it is an odd signal

$\Rightarrow$ Zero average doesn't mean RMS zero

$$\begin{aligned} \Rightarrow x_{\text{RMS}} &= \sqrt{\frac{1}{4} \left[\int_{-1}^0 1^2 dt + \int_0^1 1^2 dt \right]} \\ &= \sqrt{\frac{1}{4} [2]} \end{aligned}$$

$$x_{\text{RMS}} = \sqrt{\frac{1}{2}} \neq 0$$

Question C1:

Answer:

Given

$$k(t) = \cos(8t) + \sin(12t)$$

$$T_1 = \frac{2\pi}{8} = \frac{\pi}{4}$$

$$T_2 = \frac{2\pi}{12} = \frac{\pi}{6}$$

Now frequency ratio:

$$\frac{8}{12} = \frac{2}{3} \quad (\text{rational so periodic})$$

$$\omega_{\text{smallest}} = \text{gcd}(8, 12) = 4$$

So,

$$T_0 = \frac{2\pi}{\omega_{\text{smallest}}}$$

$$T_0 = \frac{2\pi}{4} = \frac{\pi}{2}$$

Question C2:

Answer: Given

$$x(t) = \begin{cases} \frac{2t}{\text{ms}}, & 0 \leq t < 1 \text{ ms} \\ 2, & 1 \text{ ms} \leq t < 2 \text{ ms} \\ \frac{2(3 \text{ ms} - t)}{\text{ms}}, & 2 \text{ ms} \leq t < 3 \text{ ms} \end{cases}$$

$$T_0 = 6 \text{ ms}$$

$$x(t + 3 \text{ ms}) = -x(t)$$

01.

$$X_{\text{avg}} = \frac{1}{T_0} \int_0^{T_0} x(t) dt$$

Since it is half wave symmetry,

$$\int_3^6 x(t) dt = - \int_0^3 x(t) dt$$

$$\Rightarrow \int_0^6 x(t) dt = 0$$

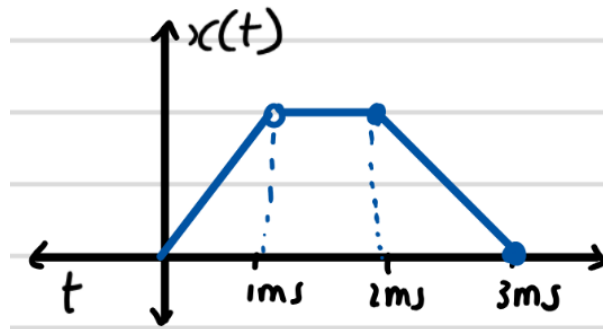


Figure 5: GraphC2

$$X_{\text{avg}} = 0$$

02.

$$|X| = \frac{1}{T_0} \int_0^{T_0} |x(t)| dt$$

$$= \frac{1}{6} \int_0^6 |x(t)| dt$$

$$= \frac{2}{6} \int_0^3 x(t) dt$$

$$= \frac{2}{6} [4]$$

$$|X| = \frac{4}{3}$$

03. RMS Value

$$X_{\text{rms}} = \sqrt{\frac{1}{T_0} \int_0^{T_0} x^2(t) dt}$$

$$x(t+3) = -x(t)$$

$$x^2(t+3) = x^2(t)$$

$$X_{\text{rms}}^2 = \frac{2}{6} \int_0^3 x^2(t) dt$$

$$1. x(t) = 2t, \quad 0 \leq t \leq 1$$

$$\Rightarrow x^2(t) = 4t^2$$

$$\Rightarrow \int_0^1 4t^2 dt = \frac{4}{3}$$

$$2. x(t) = 2, \quad 1 \leq t < 2$$

$$\Rightarrow \int_1^2 4 dt = 4$$

$$3. x(t) = 2(3 - t), \quad 2 \leq t \leq 3$$

$$\Rightarrow x^2(t) = 4(3 - t)^2$$

$$\Rightarrow \int_2^3 4(3 - t)^2 dt = \frac{4}{3}$$

$$\Rightarrow X_{\text{rms}}^2 = \frac{2}{6} \left[\frac{4}{3} + 4 + \frac{4}{3} \right]$$

$$X_{\text{rms}}^2 = \frac{20}{9}$$

$$X_{\text{rms}} = \frac{\sqrt{20}}{3}$$

$$4. P_{\text{avg}} = X_{\text{rms}}^2$$

$$P_{\text{avg}} = \frac{20}{9}$$

Question C3:

A periodic signal has half-wave symmetry and finite average power $P_x = 9$. Determine the RMS value. Now define

$$y(t) = 2x(t) + 5.$$

Derive P_y in terms of P_x and the average value of $x(t)$, and evaluate it. Explain why blindly applying

$$(2\sqrt{P_x} + 5)^2$$

is generally invalid.

Answer:

C3: Given

$$P_x = 9$$

$$y(t) = 2x(t) + 5$$

01. Average power of a signal is given by

$$P_x = x_{\text{rms}}^2$$

Given $P_x = 9$

$$9 = x_{\text{rms}}^2$$

$$\Rightarrow x_{\text{rms}} = 3$$

02.

$$y(t) = 2x(t) + 5$$

$$P_y = \langle y^2(t) \rangle$$

$$= \langle (2x(t) + 5)^2 \rangle$$

$$= \langle 4x^2(t) + 20x(t) + 25 \rangle$$

$$= 4\langle x^2(t) \rangle + 20\langle x(t) \rangle + 25$$

W.K.T. $\langle x^2(t) \rangle = P_x$

$$= 4P_x + 20\langle x(t) \rangle + 25$$

03. Since the signal is half wave symmetry,

$$x\left(t + \frac{T}{2}\right) = -x(t)$$

$\Rightarrow \langle x(t) \rangle$ for a half wave symmetry is zero

$$\langle x(t) \rangle = 0$$

So,

$$P_y = 4P_x + 20(0) + 25$$

$$= 36 + 25$$

$$\Rightarrow P_y = 61$$

04.

$$y_{\text{rms}}^2 = P_y$$

$$y_{\text{rms}} = \sqrt{P_y}$$

$$y_{\text{rms}} = \sqrt{61}$$

05. Why $(2\sqrt{P_x} + 5)^2$ wrong

Sol: RMS of a sum cannot generally be obtained by adding individual RMS values.

$$\text{Correct: } P_y = 4P_x + 20(0) + 25 = 61$$

$$\text{Wrong: } P_y = (2\sqrt{P_x} + 5)^2 = 121$$

Question D1:

Answer:

D1. a)

$$k(t) = e^{-2|t|}$$

$$\begin{aligned} E_1 &= \int_{-\infty}^{\infty} |k(t)|^2 dt \\ &= \int_{-\infty}^{\infty} e^{-4|t|} dt \end{aligned}$$

Since it is an even function,

$$\begin{aligned} &= 2 \int_0^{\infty} e^{-4t} dt \\ &= 2 \left[-\frac{1}{4} e^{-4t} \right]_0^{\infty} \\ &= -\frac{1}{2} [0 - 1] \\ &= \frac{1}{2} \end{aligned}$$

$\therefore$ It is an energy signal, the power is zero.

b)

$$\begin{aligned} k_2(t) &= e^{-t}u(t) + e^t u(-t) \\ &= e^{-|t|} \end{aligned}$$

$$\begin{aligned}
E_2 &= \int_{-\infty}^{\infty} e^{-2|t|} dt \\
&= 2 \int_0^{\infty} e^{-2t} dt \quad (\text{even function}) \\
&= 2 \left[-\frac{1}{2} e^{-2t} \right]_0^{\infty} \\
E_2 &= 1
\end{aligned}$$

$\therefore$ It is an energy signal.

c)

$$x_3(t) = 3 + 2 \cos(\omega t)$$

This is a periodic signal so a power signal.

$$\begin{aligned}
P_3 &= \langle (3 + 2 \cos(\omega t))^2 \rangle \\
&= \langle 9 + 12 \cos(\omega t) + 4 \cos^2(\omega t) \rangle \\
&= 9 + 12 \langle \cos(\omega t) \rangle + 4 \langle \cos^2(\omega t) \rangle
\end{aligned}$$

W.K.T. for one period,

$$\langle \cos(\omega t) \rangle = 0$$

$$\langle \cos^2(\omega t) \rangle = \frac{1}{2}$$

$$P_3 = 9 + 0 + 4 \left(\frac{1}{2} \right)$$

$$P_3 = 11 \quad \therefore \text{Power signal.}$$

d)

$$x_4(t) = u(t)$$

$$x_4(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$\begin{aligned}
E_4 &= \int_{-\infty}^{\infty} |x_4(t)|^2 dt \\
&= \int_0^{\infty} 1^2 dt
\end{aligned}$$

$$= \infty$$

Now,

$$\begin{aligned} P_4 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |u(t)|^2 dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^T 1 dt \\ &= \lim_{T \rightarrow \infty} \frac{T}{2T} \\ P_4 &= \frac{1}{2} \end{aligned}$$

$\therefore$ Neither Power nor energy signal

e)

$$\begin{aligned} x_5 &= \frac{\sin t}{t} \\ E_5 &= \int_{-\infty}^{\infty} \left(\frac{\sin t}{t} \right)^2 dt \end{aligned}$$

Question D2:

Answer: D2: Given

$$\begin{aligned} \phi_1(t) &= \text{rect}\left(\frac{t}{2}\right) \\ \phi_2(t) &= \text{sgn}(t) \text{rect}\left(\frac{t}{2}\right) \\ \phi_3(t) &= \left(t^2 - \frac{1}{3}\right) \text{rect}\left(\frac{t}{2}\right) \end{aligned}$$

W.K.T.

$$\langle \phi_i, \phi_j \rangle = \int_{-\infty}^{\infty} \phi_i(t) \phi_j(t) dt$$

01. W.K.T.

$$\text{rect}\left(\frac{t}{2}\right) = \begin{cases} 1, & |t| < 1 \\ 0, & \text{otherwise} \end{cases}$$

In the function $T = 2$

So $|t| < \frac{2}{2} = 1$

$$-1 < t < 1$$

and the $\text{rect}(t)$ is zero outside.

$$\text{rect}\left(\frac{t}{2}\right) = \begin{cases} 1, & |t| < 1 \\ 0, & \text{otherwise} \end{cases}$$

So for $-1 < t < 1$, rect is 1.

$$\langle \phi_1, \phi_2 \rangle = \int_{-\infty}^{\infty} \text{sgn}(t) \text{rect}^2\left(\frac{t}{2}\right) dt$$

$$= 0 \quad \text{sgn}(t) \text{ is odd \& we have symmetric limits.}$$

$$\phi_2(t) = \text{sgn}(t) \quad \text{is odd}$$

$$\phi_3(t) = t^2 - \frac{1}{3} \quad \text{is even}$$

$$\text{odd} \times \text{even} = \text{odd}$$

$$\text{odd} \times \text{odd} = \text{even}$$

$$\text{odd} \times \text{symmetric} = 0$$

02.

$$\begin{aligned} \langle \phi_1, \phi_3 \rangle &= \int_{-1}^1 \left(t^2 - \frac{1}{3} \right) dt \\ &= \left[\frac{t^3}{3} - \frac{t}{3} \right]_{-1}^1 \\ &= 0 \end{aligned}$$

$$\langle \phi_2, \phi_3 \rangle = 0$$

So ϕ_i and ϕ_j are orthogonal to each other.

$$E_1 = \langle \phi_1, \phi_1 \rangle = \int_{-1}^1 1 dt = 2$$

$$E_2 = \langle \phi_2, \phi_2 \rangle = \int_{-1}^1 1 dt = 2$$

$$E_3 = \int_{-1}^1 \left(t^2 - \frac{1}{3} \right)^2 dt$$

$$\begin{aligned}
&= 2 \int_0^1 \left(t^2 - \frac{1}{3} \right)^2 dt \\
&= 2 \left[\frac{t^5}{5} - \frac{2t^3}{9} + \frac{t}{9} \right]_0^1 \\
&= 2 \left[\frac{1}{5} - \frac{2}{9} + \frac{1}{9} \right] \\
&= \frac{8}{45}
\end{aligned}$$

Using orthogonality,

$$\begin{aligned}
E &= E_1 + E_2 + E_3 \\
&= 2 + 2 + \frac{8}{45}
\end{aligned}$$

Question D3:

Answer: D3: Given

$$x_a(t) = e^{-a|t|}, \quad a > 0$$

01.

$$\lim_{t \rightarrow \infty} e^{-a|t|} = e^{-\infty} = 0$$

When $t \rightarrow 0^+$,

$$\lim_{t \rightarrow 0^+} e^{-a|t|} = e^0 = 1$$

02.

$$E_{x_a} = \int_{-\infty}^{\infty} |x_a(t)|^2 dt$$

$$x_a(t) = e^{-a|t|} \Rightarrow E_{x_a} = \int_{-\infty}^{\infty} e^{-2a|t|} dt$$

Since it is an even function,

$$\begin{aligned}
E_{x_a} &= 2 \int_0^{\infty} e^{-2at} dt \\
&= 2 \left[-\frac{1}{2a} e^{-2at} \right]_0^{\infty}
\end{aligned}$$

$$E_{x_a} = \frac{1}{a}$$

$\therefore x_a(t)$ is an energy signal for every value of $a > 0$.

03.

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x_a(t)|^2 dt$$

Now,

$$\begin{aligned} P_x &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T e^{-2a|t|} dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T e^{-2at} dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \left[-\frac{1}{2a} e^{-2at} \right]_0^T \\ &= \lim_{T \rightarrow \infty} \frac{1}{2aT} (1 - e^{-2aT}) \end{aligned}$$

$$P_x = 0$$

04.

$$x(t) = 1$$

$$E_x = \int_{-\infty}^{\infty} 1 dt = \infty$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T 1 dt = 1$$

Question D4:

Answer: D4: Given

$$E_x = 4, \quad E_y = 9, \quad \langle x, y \rangle = -1$$

$$E_{x+\alpha y} = E_x + \alpha^2 E_y + 2\alpha \langle x, y \rangle$$

$$= 4 + 9\alpha^2 - 2\alpha$$

01.

$$E_{x+y} = E_x + E_y + 2\langle x, y \rangle$$

$$= 4 + 9 + 2(-1) = 11$$

02.

$$E_{2x-y} = 4E_x + E_y - 4\langle x, y \rangle$$

$$= 4(4) + 9 - 4(-1) = 29$$

03.

$$E_{x-y} = E_x + E_y - 2\langle x, y \rangle$$

$$= 4 + 9 - 2(-1)$$

$$= 15$$

04.

$$\langle x + \alpha y, y \rangle = 0$$

$$\langle x, y \rangle + \alpha \langle y, y \rangle = 0$$

$$-1 + 9\alpha = 0$$

$$\Rightarrow \alpha = \frac{1}{9}$$

$$E_{x+\alpha y} = 4 + 9\alpha^2 - 2\alpha$$

$$\alpha = \frac{1}{9}$$

$$\begin{aligned} E_{\min} &= 4 + 9 \left[\frac{1}{9} \right]^2 - 2 \left[\frac{1}{9} \right] \\ &= \frac{35}{9} \end{aligned}$$

Question E1:

Answer:

$$s(t) = r(t+2) - 3r(t+1) + 3r(t) - r(t-1)$$

$$r(t) = tu(t)$$

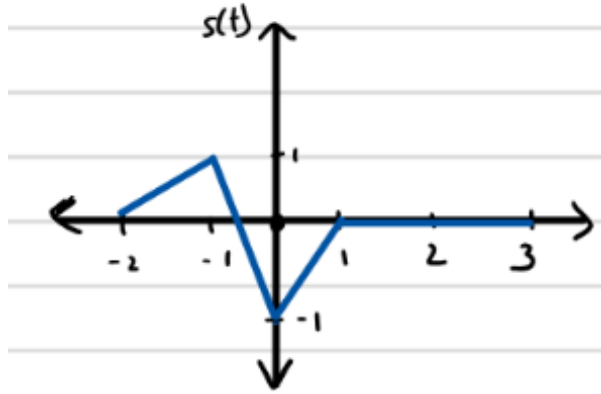


Figure 6: GraphE1

01.

$$r(t+2) = (t+2)u(t+2)$$

$$r(t+1) = (t+1)u(t+1)$$

$$r(t) = tu(t)$$

$$r(t-1) = (t-1)u(t-1)$$

$$s(t) = (t+2)u(t+2) - 3(t+1)u(t+1)$$

$$+ 3tu(t) - (t-1)u(t-1)$$

Now,

$$s(t) = \begin{cases} 0, & t < -2 \\ t+2, & -2 \leq t < -1 \\ -2t-1, & -1 \leq t < 0 \\ t-1, & 0 \leq t < 1 \\ 0, & t \geq 1 \end{cases}$$

02.

$$\frac{ds}{dt} = u(t+2) - 3u(t+1) + 3u(t) - u(t-1)$$

03.

$$\frac{d^2s}{dt^2} = \delta(t+2) - 3\delta(t+1) + 3\delta(t) - \delta(t-1)$$

04. $\frac{d^3 s}{dt^3}$ (third derivative)

$$\frac{d^3 s}{dt^3} = \delta'(t+2) - 3\delta'(t+1) + 3\delta'(t) - \delta'(t-1)$$

Question E2:

Answer:

E2: Given

$$x(t) = \begin{cases} 0, & t < -3 \\ t+3, & -3 \leq t < -1 \\ 2, & -1 \leq t < 1 \\ 3-t, & 1 \leq t < 3 \\ 0, & t \geq 3 \end{cases}$$

01. W.K.T.

$$r(t) = tu(t)$$

$$\frac{d}{dt}r(t) = u(t)$$

$$\frac{d}{dt}u(t) = \delta(t)$$

02.

$$x(t) = r(t+3) - r(t+1) - r(t-1) + r(t-3)$$

03.

$$\frac{dx(t)}{dt} = u(t+3) - u(t+1) - u(t-1) + u(t-3)$$

$$x(t) = r(t+3) - r(t+1) - r(t-1) + r(t-3)$$

$$x'(t) = u(t+3) - u(t+1) - u(t-1) + u(t-3)$$

$$x''(t) = \delta(t+3) - \delta(t+1) - \delta(t-1) + \delta(t-3)$$

Question E3:

Answer:

E3: Given

$$x(t) = \text{rect}\left(\frac{t-1}{4}\right) + \text{rect}\left(\frac{t+1}{2}\right) - 2\text{rect}(t-1)$$

01. We know that

$$\text{rect}\left(\frac{t-a}{T}\right) = 1, \quad a - \frac{T}{2} < t < a + \frac{T}{2}$$

$$01. \text{rect}\left(\frac{t-1}{4}\right) = 1, \quad -1 < t < 3$$

$$02. \text{rect}\left(\frac{t+1}{2}\right) = 1, \quad -2 < t < 0$$

$$03. \text{rect}(t-1) = 1, \quad \frac{1}{2} < t < \frac{3}{2}$$

02.

$$\begin{aligned} t < -2 & \quad x(t) = 0 \\ -2 < t < -1 & \quad x(t) = 1 \\ -1 < t < 0 & \quad x(t) = 1 + 1 = 2 \\ 0 < t < \frac{1}{2} & \quad x(t) = 1 \\ \frac{1}{2} < t < \frac{3}{2} & \quad x(t) = 1 - 2 = -1 \\ \frac{3}{2} < t < 3 & \quad x(t) = 1 \\ t > 3 & \quad x(t) = 0 \end{aligned}$$

$$x(t) = \begin{cases} 0, & t < -2 \\ 1, & -2 < t < -1 \\ 2, & -1 < t < 0 \\ 1, & 0 < t < \frac{1}{2} \\ -1, & \frac{1}{2} < t < \frac{3}{2} \\ 1, & \frac{3}{2} < t < 3 \\ 0, & t > 3 \end{cases}$$

03.

$$\begin{aligned} E_x &= \int_{-\infty}^{\infty} |x(t)|^2 dt \\ &= 1^2(1) + 2^2(1) + 1^2\left(\frac{1}{2}\right) + (-1)^2(1) + 1^2\left(\frac{3}{2}\right) \end{aligned}$$

$$E_x = 8$$

04.

$$P_x = 0 \quad \text{since } E_x \text{ is finite}$$

05.

$$x_0(t) = \frac{1}{2} \left[\text{rect} \left(\frac{t-1}{4} \right) + \text{rect} \left(\frac{t+1}{2} \right) - 2 \text{rect}(t-1) \right] \\ + \text{rect} \left(\frac{t+1}{4} \right) + \text{rect} \left(\frac{t-1}{2} \right) - 2 \text{rect}(t+1)$$

Question E4:

Answer:

E4: Given

$$I_1 = \int_{-\infty}^{\infty} (t^2 - 3t + 1) \delta(t-2) dt$$

$$I_2 = \int_{-\infty}^{\infty} x(t) \delta(2t+4) dt$$

$$I_3 = \int_{-\infty}^{\infty} \delta(t+4) u(2-t) dt$$

$$x(t) = e^{-t} u(t) + 2u(-t)$$

01.

$$I_1 = 1$$

02.

$$I_2 = \int_{-\infty}^{\infty} x(t) \delta(2t+4) dt$$

$$\delta(2t+4) = \frac{1}{|2|} \delta(t+2)$$

$$\text{So } \delta(2(t+2)) = \frac{1}{2} \delta(t+2)$$

$$\Rightarrow \int_{-\infty}^{\infty} x(t) \frac{1}{2} \delta(t+2) dt$$

$$I_2 = \frac{1}{2} x(-2)$$

$$\Rightarrow x(-2) = e^{-(-2)} u(-2) + 2u(2)$$

$$= 0 + 2$$

$$\Rightarrow I_2 = \frac{1}{2}(2) = 1$$

03.

$$I_3 = \int_{-\infty}^{\infty} \delta(t+4)u(2-t) dt$$

$$\int f(t)\delta(t-a) dt = f(a)$$

$$\Rightarrow I_3 = u(2 - (-4))$$

$$\Rightarrow I_3 = u(6) = 1$$

04. Used formulae

$$01. \delta(at+b) = \frac{1}{|a|}\delta\left(t + \frac{b}{a}\right)$$

$$02. \delta(2t+4) = \frac{1}{2}\delta(t+2)$$

$$03. \int f(t)\delta(t-a) dt = f(a)$$

05. Hence

$$I_1 = -1$$

$$I_2 = 1$$

$$I_3 = 1$$